

18. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.

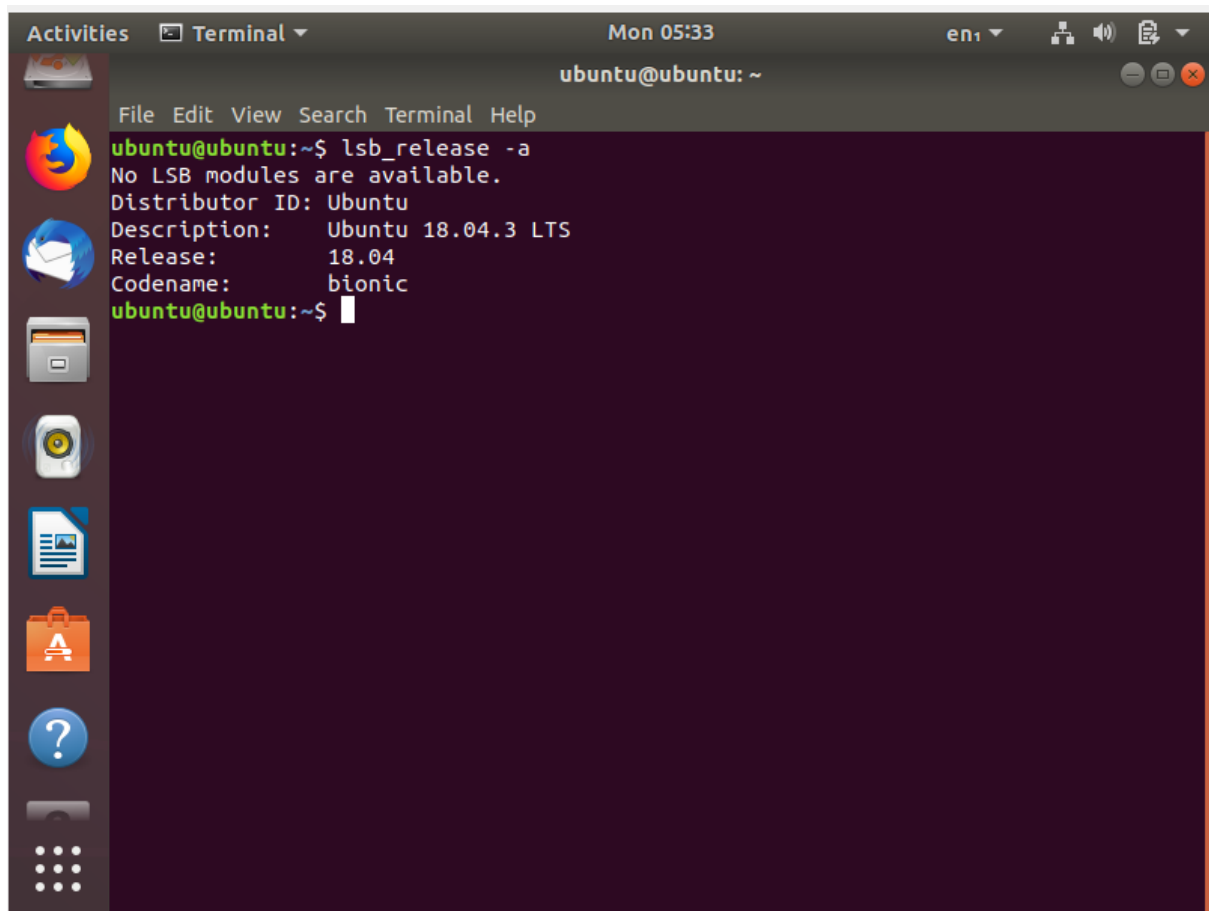
AIM

To install and configure Oracle VM VirtualBox on a Windows host and create a virtual machine with a Linux/Windows operating system running on top of the existing Windows environment.

PROCEDURE

- Install Oracle VM VirtualBox on the Windows computer.
- Download the required operating-system ISO file, such as Ubuntu Linux or Windows. VirtualBox requires an OS installation image such as an ISO.
- Oracle Documentation
- Open Oracle VM VirtualBox Manager.
- Click New to create a virtual machine.
- Oracle Documentation
- Enter a name for the VM, for example:
 - Ubuntu VM
 - Windows VM
- Select the appropriate OS Type and Version.
- Select the downloaded ISO Image.
- Allocate RAM and CPU resources to the virtual machine.
- Create a virtual hard disk and specify its storage size.
- Click Finish to create the VM.
- Select the VM and click Start.
- The VM boots from the selected ISO and starts the operating-system installation.
- Oracle Documentation
- Follow the OS installation instructions and create the required username and password.
- After installation, restart the VM and log in to the installed operating system.
- The guest OS can now be used independently inside the Windows host.

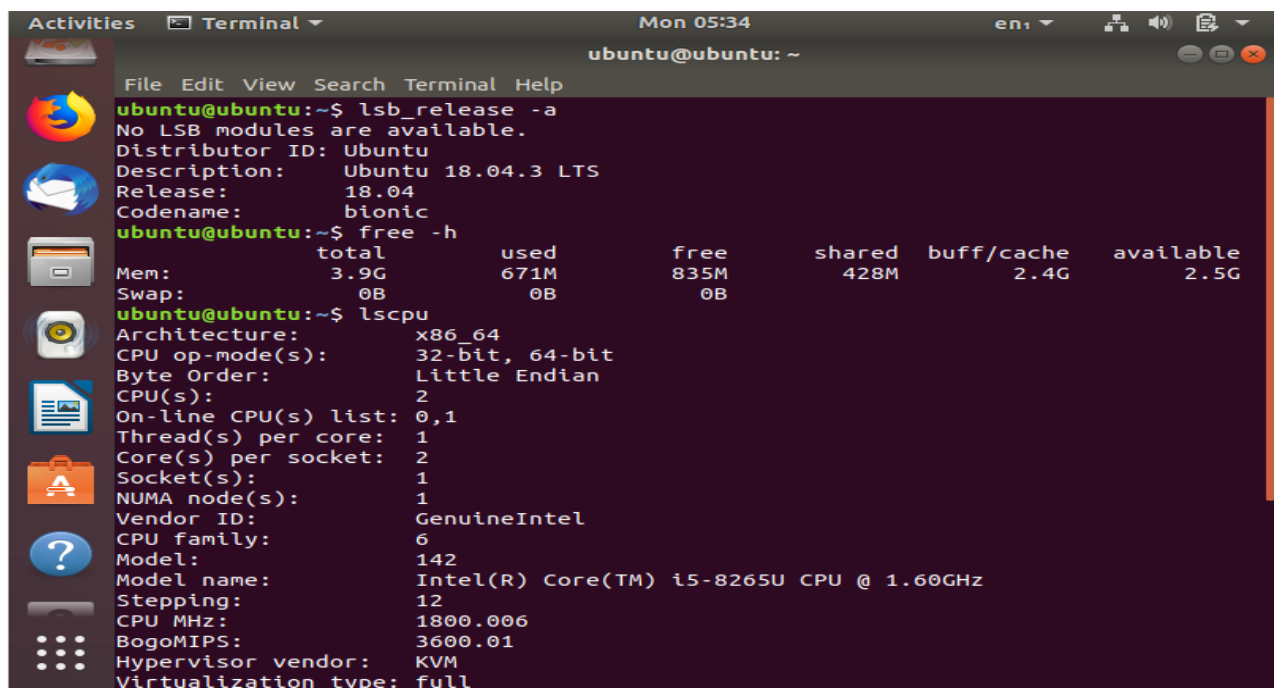
VIRTUALBOX MANAGER



A terminal window titled 'Terminal' with a menu bar (File, Edit, View, Search, Terminal, Help) and a status bar (Mon 05:33, en1). The prompt is 'ubuntu@ubuntu: ~'. The command 'lsb_release -a' has been executed, showing the following output:

```
ubuntu@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 18.04.3 LTS
Release:        18.04
Codename:       bionic
ubuntu@ubuntu:~$
```

UBUNTU VERSION



A terminal window titled 'Terminal' with a menu bar (File, Edit, View, Search, Terminal, Help) and a status bar (Mon 05:34, en1). The prompt is 'ubuntu@ubuntu: ~'. The command 'lsb_release -a' has been executed, showing the same output as the previous image. Below that, the command 'free -h' has been executed, showing the following output:

```
ubuntu@ubuntu:~$ free -h
              total        used        free      shared    buff/cache   available
Mem:           3.9G          671M          835M          428M          2.4G          2.5G
Swap:            0B           0B           0B
```

Then, the command 'lscpu' has been executed, showing the following output:

```
ubuntu@ubuntu:~$ lscpu
Architecture:        x86_64
CPU op-mode(s):      32-bit, 64-bit
Byte Order:          Little Endian
CPU(s):               2
On-line CPU(s) list: 0,1
Thread(s) per core:   1
Core(s) per socket:   2
Socket(s):            1
NUMA node(s):         1
Vendor ID:            GenuineIntel
CPU family:           6
Model:               142
Model name:           Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz
Stepping:             12
CPU MHz:              1800.006
BogoMIPS:             3600.01
Hypervisor vendor:    KVM
Virtualization type:  full
```

```
Activities Terminal Mon 05:35 ubuntu@ubuntu: ~
File Edit View Search Terminal Help
Swap: 0B 0B 0B
ubuntu@ubuntu:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
NUMA node(s): 1
Vendor ID: GenuineIntel
CPU family: 6
Model: 142
Model name: Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz
Stepping: 12
CPU MHz: 1800.006
BogoMIPS: 3600.01
Hypervisor vendor: KVM
Virtualization type: full
L1d cache: 32K
L1i cache: 32K
L2 cache: 256K
L3 cache: 6144K
NUMA node0 CPU(s): 0,1
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca c
mov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc re
p_good nopl xtology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 cx16
pcid sse4_1 sse4_2 movbe popcnt aes rdrand hypervisor lahf_lm abm 3dnowprefetch
```

NETWORK TEST

```
Activities Terminal Mon 05:36 ubuntu@ubuntu: ~
File Edit View Search Terminal Help
tmpfs 2.0G 0 2.0G 0% /dev/shm
tmpfs 5.0M 8.0K 5.0M 1% /run/lock
tmpfs 2.0G 0 2.0G 0% /sys/fs/cgroup
tmpfs 2.0G 0 2.0G 0% /tmp
tmpfs 395M 60K 395M 1% /run/user/999
/dev/loop1 89M 89M 0 100% /snap/core/7270
/dev/loop2 55M 55M 0 100% /snap/core18/1066
/dev/loop3 43M 43M 0 100% /snap/gtk-common-themes/1313
/dev/loop4 150M 150M 0 100% /snap/gnome-3-28-1804/67
/dev/loop5 4.2M 4.2M 0 100% /snap/gnome-calculator/406
/dev/loop6 15M 15M 0 100% /snap/gnome-characters/296
/dev/loop7 1.0M 1.0M 0 100% /snap/gnome-logs/61
/dev/loop8 3.8M 3.8M 0 100% /snap/gnome-system-monitor/100
ubuntu@ubuntu:~$ ping -c 4 google.com
PING google.com (142.250.205.110) 56(84) bytes of data:
64 bytes from pnmaaa-ao-in-f14.1e100.net (142.250.205.110): icmp_seq=1 ttl=116
time=13.8 ms
64 bytes from pnmaaa-ao-in-f14.1e100.net (142.250.205.110): icmp_seq=2 ttl=116
time=12.5 ms
64 bytes from pnmaaa-ao-in-f14.1e100.net (142.250.205.110): icmp_seq=3 ttl=116
time=34.6 ms
64 bytes from pnmaaa-ao-in-f14.1e100.net (142.250.205.110): icmp_seq=4 ttl=116
time=13.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3017ms
rtt min/avg/max/mdev = 12.552/18.635/34.626/9.245 ms
ubuntu@ubuntu:~$
```

RESULT

The Oracle VM VirtualBox was successfully installed and configured, and a Linux/Windows operating system was successfully installed and executed as a virtual machine on top of the existing Windows environment.